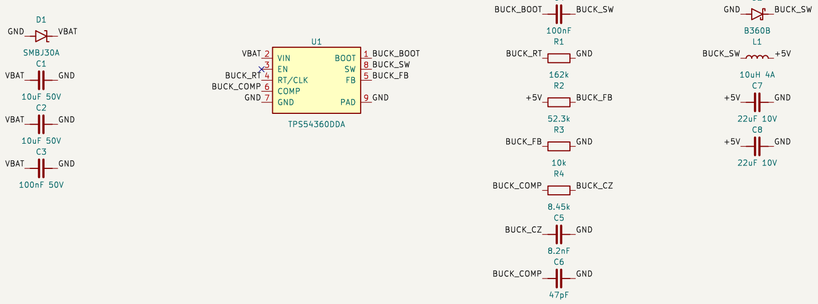
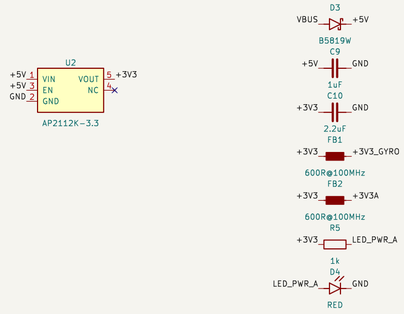
1. BATTERY INPUT + 5 V BUCK CONVERTER

Battery arrives through the ESC cable. TVS + caps tame spikes; the TPS54360 makes 5 V.



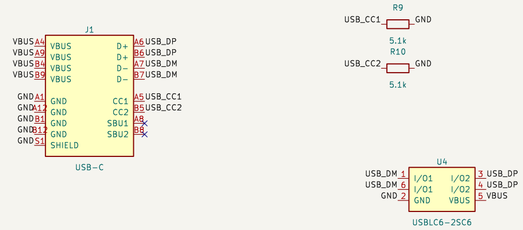
2. USB POWER PATH + 3.3 V LDO + FILTERED RAILS

USB can power the board alone (no motors). The LDO makes 3.3 V; beads give the gyro and ADC quiet copies.



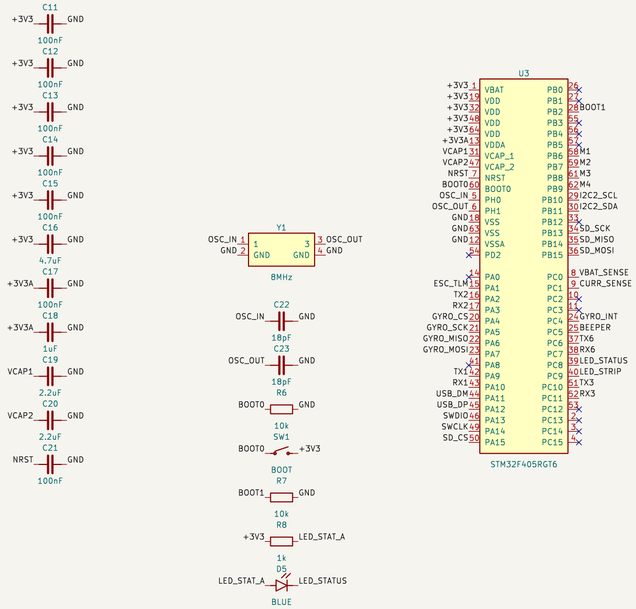
4. USB-C + ESD

CC resistors make chargers and laptops supply 5 V. USBLC6 protects D+/D− from static.



3. MCU: STM32F405RGT6

Pin map copied from the SpeedyBee F405 V3 Betaflight target. Unused pins carry an X (no-connect).

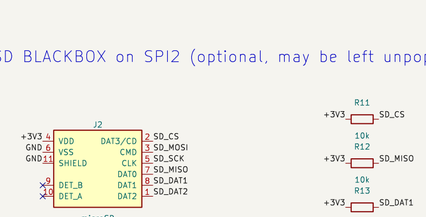


5. GYRO: BMI270 on SPI1

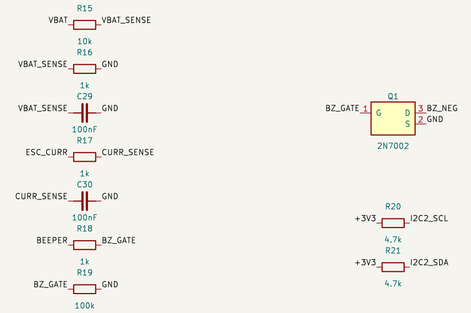
Fed from the filtered +3V3_GYRO rail. Aux pins unused (X).



6. microSD BLACKBOX on SPI2 (optional, may be left unpopulated)

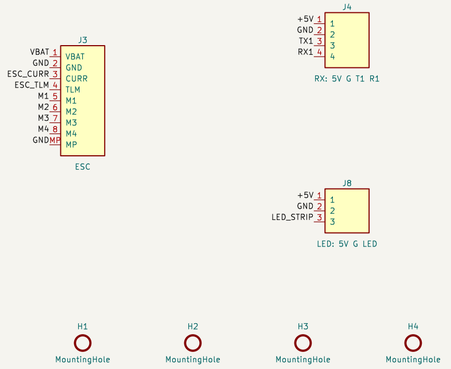


7. BATTERY/CURRENT SENSE, BEEPER, I2C PULL-UPS



8. CONNECTORS AND SOLDER PADS

ESC plug pin order is NOT standard: match it to your ESC's cable before ordering.



POWER FLAGS (tell ERC where power comes from)



HOW TO READ THIS SCHEMATIC: there are almost no wires. A net label touching a pin connects it; every label with the same name is the same wire.

36 x 36 mm, 30.5 mm mounting, 2S–6S
Betaflight target: SPEEDYBEEF405V3 (pin-compatible)
Kathmandu University Robotics Club

Sheet: /
File: kurc_f405.kicad_sch

Title: KURC F405 Flight Controller

Size: A2	Date: 2026-10-30	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1